

ICML-Style Final Report: Resume–Job Matching with TF-IDF, SBERT, and LLM-Assisted Reranking

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Abstract

We study resume–job matching as a ranking problem using a reproducible pipeline built from public resume and job-posting data. Our dataset contains 1,309 processed resumes, 2,119 processed jobs, and 10,472 weakly labeled resume–job pairs across ten normalized career categories. We compare a TF-IDF lexical retriever, an SBERT dense retriever, and a Gemma-based LLM reranker applied to a lexical shortlist. On the full held-out test split of 823 jobs, TF-IDF remains the strongest model overall, achieving Precision@10 of 0.039 and NDCG@10 of 0.132. The LLM reranker is second-best on the same full split, with Precision@10 of 0.037 and NDCG@10 of 0.125, while SBERT trails both alternatives at 0.029 and 0.100. Bootstrap testing shows the LLM significantly outperforms SBERT on the full test split, but TF-IDF retains a small advantage over the LLM on Precision@10 and NDCG@10. Interpretability evidence is strongest for TF-IDF and SBERT, while the full-test LLM run uses compact prompts that reduce explanation richness. Fairness and robustness stress tests show TF-IDF is most stable under demographic perturbations, SBERT is notably more sensitive to name swaps, and the LLM fairness evidence remains too limited for a strong quantitative claim. The main business implication is that a strong lexical baseline remains hard to beat under limited, weakly supervised data, while LLM reranking is promising but still expensive and only partially validated for deployment.

1 Introduction and Problem Motivation

Organizations often receive many resumes per vacancy, making manual first-pass screening slow, inconsistent, and costly. A ranking system that surfaces more relevant resumes earlier can shorten time-to-screen, reduce recruiter workload, and improve candidate experience. In

business settings, however, a useful system must satisfy more than raw ranking quality: it should be reproducible, interpretable enough for audit, and robust to non-job-related textual changes.

This project frames resume screening as a *resume retrieval and ranking* task. Each job posting is treated as a query, and resumes are ranked by estimated fit. We compare three families of methods with increasing modeling complexity:

- **TF-IDF lexical matching**, as a simple and transparent baseline.
- **SBERT dense matching**, to test whether semantic embeddings improve generalization beyond exact term overlap.
- **LLM-assisted reranking**, where a local Gemma model reranks a short lexical shortlist.

The practical question is not merely which method scores highest on one subset, but which method best balances quality, speed, interpretability, and deployment risk for a resume-screening workflow.

2 Related Work Context

TF-IDF-style sparse retrieval remains a durable baseline in information retrieval because it is simple, data-efficient, and highly interpretable [?]. Dense sentence embeddings such as SBERT are often preferred when semantic similarity matters more than literal token overlap [?]. For ranking evaluation, graded metrics such as NDCG [?] and early-hit metrics such as reciprocal rank [?] are standard. This project situates resume screening within that retrieval tradition while also adding interpretability and fairness diagnostics.

In hiring-related analytics, the technical objective cannot be separated from governance concerns. Demographic proxies in text may affect rankings, and natural-language explanations can appear convincing even when the un-

Table 1: Dataset facts from the generated preprocessing artifacts.

Asset / statistic	Value
Processed resumes	1,309
Processed jobs	2,119
Constructed pairs	10,472
Test query jobs	823
Test resume candidates	197
Mean resume length (processed chars)	5,971
Mean job length (processed chars)	2,157
Label 0 / 1 / 2 counts	3927 / 2618 / 3927

derlying evidence is weak. That is why the project does not stop at point metrics and instead compares methods across quality, robustness, and operational cost.

3 Dataset and Preprocessing

3.1 Source data and constructed benchmark

The project combines a public resume dataset and a public fake-job-postings dataset, filtering the latter to non-fraudulent jobs. After cleaning and category normalization, the pipeline produces 1,309 resumes, 2,119 jobs, and 10,472 resume–job pairs. The data span ten normalized categories: accounting/finance, creative/design/media, education/training, healthcare, hospitality/wellness, HR/customer support, legal/consulting, operations/engineering, sales/marketing, and technology/data.

The benchmark is weakly supervised. For each resume, the preprocessing pipeline constructs:

- 3 **positive** pairs (label 2) from the same normalized category with strongest overlap,
- 2 **partial** pairs (label 1) from related categories,
- 3 **negative** pairs (label 0) from distant categories.

This yields a balanced label distribution for labels 0 and 2, with fewer partial matches: 3,927 negatives, 2,618 partial matches, and 3,927 strong matches. The final split contains 7,328 training pairs, 1,568 validation pairs, and 1,576 test pairs.

3.2 Cleaning and feature extraction

The preprocessing pipeline removes HTML and URLs, normalizes whitespace, masks direct personally identifiable information such as emails and phone numbers, and optionally masks named entities with spaCy. Resume cat-

Table 2: Normalized category coverage from the generated dataset summary.

Category	Resumes	Jobs
Accounting/finance	140	220
Creative/design/media	140	220
Education/training	102	220
Healthcare	115	220
Hospitality/wellness	140	139
HR/customer support	132	220
Legal/consulting	140	220
Operations/engineering	140	220
Sales/marketing	140	220
Technology/data	120	220

egories are mapped into ten normalized sectors, while job categories are inferred using rule-based keyword matching over title, function, department, and industry fields. The pipeline also extracts a small curated skill vocabulary via regular expressions (for example Python, SQL, accounting, marketing, logistics, and cooking).

These steps make the benchmark operationally useful, but they also introduce assumptions: category normalization is coarse, regex skill extraction is incomplete, and the constructed labels reflect overlap heuristics rather than observed recruiter judgments.

4 Methods

4.1 TF-IDF baseline

Our baseline uses a TF-IDF vectorizer with English stop-word removal, sublinear TF scaling, and unigram–bigram features. Each job posting and resume is converted into a sparse vector, and cosine similarity is used for ranking. The method is attractive because it is cheap, deterministic, and easy to explain: top-ranked matches can be traced to shared terms and n-grams.

4.2 SBERT dense retrieval

The dense model uses sentence-transformers/all-MiniLM-L6-v2 [?]. Jobs and resumes are embedded into a shared dense space using normalized sentence embeddings, and rankings are produced from dot products. Embeddings are cached for speed. This model is intended to capture semantic similarity beyond literal keyword overlap, such as near-synonymous skill descriptions.

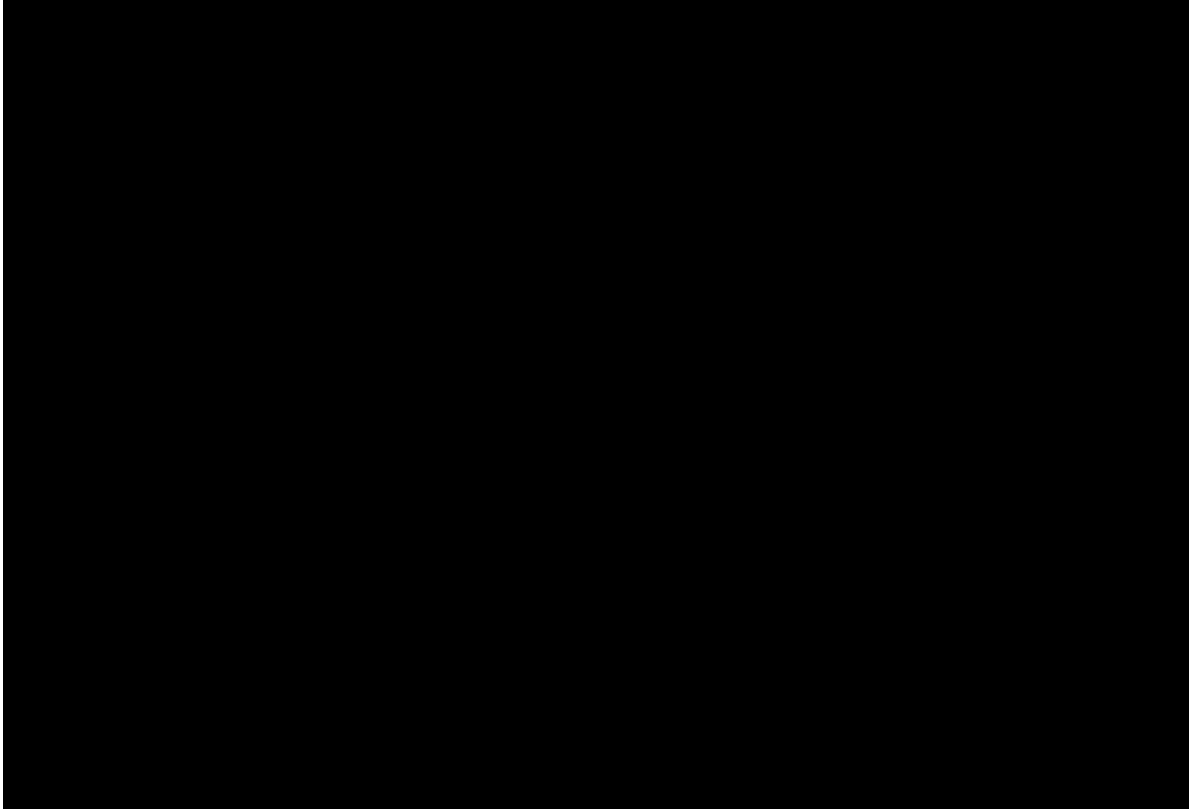


Figure 1: Generated data overview figure used in the report. It summarizes category coverage and length-related characteristics of the processed benchmark.

4.3 LLM-assisted reranking

The LLM approach does not score all resumes directly. Instead, it first builds a lexical shortlist and then asks a local Gemma model, served through LM Studio, to rerank the shortlisted resumes using structured JSON output. In the final full-test pass, the reranker still sees only three shortlisted resumes per job, but the prompt is compact: job and resume snippets are truncated more aggressively and the output is reduced to concise ranking JSON without rich free-text explanations. This keeps inference tractable enough to evaluate all 823 held-out test jobs while preserving the same two-stage ranking design.

4.4 Why these methods are complementary

The three approaches represent different deployment philosophies. TF-IDF emphasizes transparency and low cost. SBERT tests whether a denser semantic representation can recover good matches even when exact wording differs. The LLM reranker explores a two-stage system in which a cheap retriever narrows the search space and a stronger model makes the final shortlist ordering more human-like. This comparison is practically relevant be-

cause real enterprise screening systems are often hybrid rather than single-model pipelines.

5 Experimental Setup and Metrics

Each unique job in a split is treated as a query, and the unique resumes in that split form the candidate set. The evaluation pipeline computes query-level and aggregate ranking metrics:

- **Precision@10** for shortlist quality,
- **MRR** for how quickly a relevant resume appears,
- **MAP** for ranking quality over all relevant resumes,
- **NDCG@10** for graded ranking quality.

For Precision, Recall, MRR, and MAP, labels 1 and 2 are treated as relevant; NDCG retains the original graded labels. We also report paired bootstrap significance tests. Because the LLM was eventually evaluated on the full held-out test split, all pairwise full-test comparisons are now statistically comparable.

5.1 Implementation details

The full test evaluation ranks 197 unique resume candidates for each of 823 unique job queries. The TF-IDF model uses sparse unigram–bigram features with English stopword removal and cosine similarity. SBERT uses normalized `all-MiniLM-L6-v2` embeddings with cached vectors. The LLM pipeline uses a local LM Studio endpoint, requests compact structured JSON rankings for a shortlist of three resumes per job, and falls back to lexical order if parsing or generation fails. Consequently, LLM performance reflects both ranking quality and output-format reliability.

6 Results

6.1 Main comparison

Table ?? summarizes the full-test evaluation across all three model families on the same 823 held-out jobs.

On the full test split, TF-IDF remains the strongest model overall. The margin over SBERT is meaningful rather than cosmetic: Precision@10 improves from 0.0288 to 0.0391, MRR from 0.1103 to 0.1409, and NDCG@10 from 0.0999 to 0.1323. The full-test LLM run lands between TF-IDF and SBERT. It improves materially over SBERT on all three main metrics, but it does not displace TF-IDF as the best model on the full held-out benchmark. This outcome suggests that the current benchmark rewards explicit lexical and category overlap more strongly than semantic abstraction. That is plausible because the labels themselves are partly constructed from overlap heuristics. In other words, TF-IDF is well aligned to the way supervision was created.

6.2 What the full-test LLM run changes

The full-test LLM result sharpens the earlier story rather than overturning it. The LLM clearly beats SBERT on the held-out test split, and the bootstrap results confirm that these gains are statistically significant for Precision@10, MRR, and NDCG@10. However, TF-IDF still holds a small but measurable edge over the LLM on Precision@10 and NDCG@10, and the MRR difference is not statistically significant. That makes the final ranking story more complete: TF-IDF remains first, the LLM becomes a stronger second, and SBERT remains third.

The absolute metric values are still modest. Even the strongest model retrieves fewer than one labeled-relevant resume in the top ten on average under the current benchmark. This does not make the comparison useless, but it does mean the results should be read as comparative evi-

dence under weak supervision rather than as deployment-ready hiring quality.

6.3 Runtime and operating cost

Figure ?? and the saved runtime table show large deployment trade-offs. TF-IDF completes the full 823-job test run in about 3.56 seconds, SBERT completes in about 28.47 seconds, and the LLM consumes about 3,384.5 seconds for the same 823 jobs, or about 4.11 seconds per query. The full-test LLM run was made tractable by using compact prompts and concise JSON outputs, but it is still far slower than the non-LLM alternatives.

7 Interpretability Analysis

Interpretability was studied using the generated example table and model-specific attribution mechanisms.

For **TF-IDF**, explanations are straightforward. The generated examples show that top matches are driven by transparent lexical evidence such as *accounting*, *finance*, *aml*, and *kyc*. This is valuable for auditability because recruiters can inspect why a resume surfaced.

For **SBERT**, the project uses perturbation-style attribution: job or resume chunks are removed and the embedding similarity drop is measured. The resulting examples show broader evidence spans, such as multi-sentence chunks describing kitchen operations or teamwork for hospitality roles. This is richer than raw term overlap but less immediately interpretable than TF-IDF.

For the **LLM**, explanations come from shortlist rationales or overlap proxies. In principle this is the most recruiter-friendly explanation style, but the current active notebook-first artifact set does not preserve a stable LLM example row in the same way that it preserves TF-IDF and SBERT examples. That gap is itself informative: the explanation mechanism is narrower in scope and less reproducible than the classical alternatives.

Overall, TF-IDF is the most auditable, SBERT is the most indirect, and the LLM is potentially the most natural-language-friendly but also the least fully validated in the current active export set.

The interpretability artifacts also expose different failure modes. TF-IDF can miss semantically relevant evidence when wording diverges. SBERT can rely on broader context chunks that are harder to audit line-by-line. LLM explanations are readable and recruiter-friendly in principle, but the active artifact trail is thinner and therefore harder to defend rigorously. That difference matters when explanation text may influence human decision-makers.

Table 3: Main comparison table from the generated report assets. All three rows now correspond to the same held-out test split.

Model	Scope	Queries	Precision@10	MRR	NDCG@10	MAP
TF-IDF baseline	Full held-out test split	823	0.039	0.141	0.132	0.108
SBERT	Full held-out test split	823	0.029	0.110	0.100	0.083
LLM reranker	Full held-out test split	823	0.037	0.136	0.125	0.103

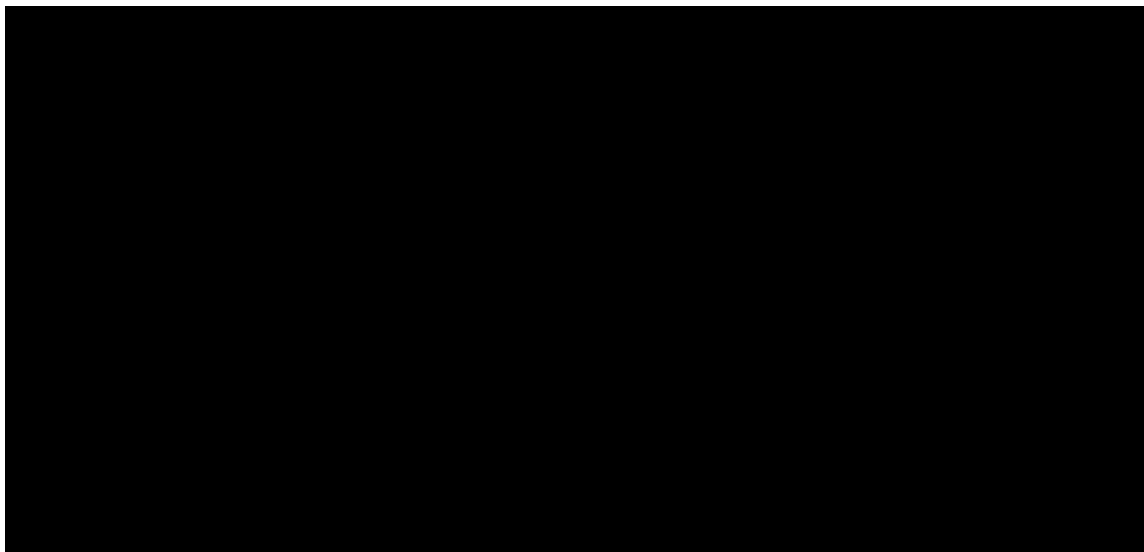


Figure 2: Generated figure comparing the main metrics across models on the same held-out test split.

Table 4: Generated bootstrap significance results. Positive deltas favor the left-hand model.

Comparison	Metric	Delta	<i>p</i>
TF-IDF SBERT	P@10	0.010	0.0000
TF-IDF SBERT	MRR	0.031	0.0004
TF-IDF SBERT	NDCG@100	0.032	0.0000
TF-IDF – LLM	P@10	0.002	0.0052
TF-IDF – LLM	MRR	0.004	0.1788
TF-IDF – LLM	NDCG@100	0.007	0.0120
SBERT – LLM	P@10	−0.008	0.0004
SBERT – LLM	MRR	−0.026	0.0024
SBERT – LLM	NDCG@100	0.025	0.0000

8 Fairness and Robustness Evaluation

The project runs perturbation-based fairness and robustness audits by modifying resumes with demographic signals such as gendered wording, ethnically coded names, and age-related information, then measuring how rank-

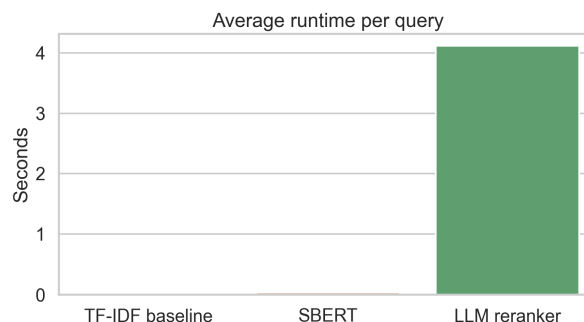


Figure 3: Generated runtime comparison figure. All three models are now measured on the same held-out test split.

ings change. The active notebook-first results support full summary tables for TF-IDF and SBERT on a 25-job validation subset. LLM fairness remains under-evaluated and is therefore discussed only as an unresolved gap rather than as a stable quantitative result.

TF-IDF is the most stable model overall. Its mean Top-1 change rate is only 2.3%, and its worst perturbation reaches 4%. SBERT is materially less robust, with a mean Top-1 change rate of 21.7% and a worst case of 44%. The detailed fairness table shows that SBERT is particularly sensitive to name swaps: East Asian, South

Table 5: Representative generated interpretability examples from the active artifact set. Text is lightly truncated for layout.

Family	Job title	Job category	Resume category	Shared skills	Evidence summary
TF-IDF	CFO/Controller	accounting_finance	accounting_finance	accounting; auditing; legal_research; payroll	Lexical attribution highlights strong overlap in accounting and finance terms, with top supporting evidence including <i>accounting</i> , <i>finance</i> , and <i>accounting finance</i> .
SBERT	Sous Chef	hospitality_wellness	hospitality_wellness	cooking; logistics; manufacturing	Chunk ablation shows the score is supported by operational kitchen text, teamwork descriptions, and experience narratives beyond isolated keywords.

Table 6: Overview of active fairness results. Lower Top-1 change and higher overlap are better.

Model	Top-1 change	Top-10 overlap	Spearman V
TF-IDF	0.023	0.994	0.999
SBERT	0.217	0.888	0.954

Table 7: Selected perturbation detail from the generated fairness table. SBERT is especially sensitive to name swaps.

Perturbation	TF-IDF Top-1	SBERT Top-1	SBERT Top-10
Age reduced	0.00	0.20	0.928
African American name	0.04	0.20	0.804
East Asian name	0.04	0.44	0.800
South Asian name	0.04	0.32	0.848
White American name	0.04	0.36	0.848
Gender feminine	0.00	0.00	1.000
Gender masculine	0.00	0.00	0.988

Asian, African American, and White American coded names all trigger much larger rank movement than the same perturbations under TF-IDF.

The active artifact set does not support a strong LLM fairness estimate. Rather than reporting a number that looks more precise than it really is, the more honest conclusion is that LLM fairness remains essentially unvalidated in the current project delivery.

Even though these perturbations do not by themselves prove real-world discrimination, they are valuable screening stress tests. A model whose top-ranked candidate changes sharply after a name swap deserves substantially more governance scrutiny before production use.

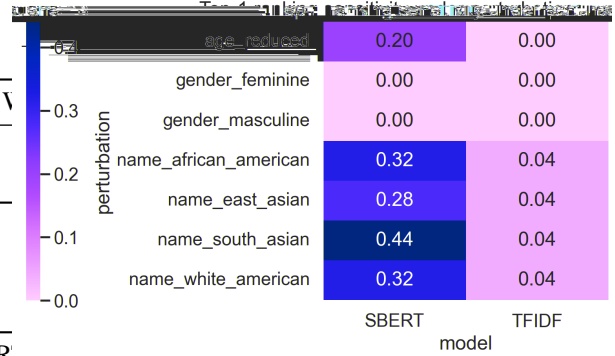


Figure 9.28: Generated heatmap of Top-1 change rates under perturbation. SBERT is visibly more sensitive than TF-IDF in the active audit.

9 Discussion, Limitations, and Business Implications

9.1 Strengths

The project has four main strengths. First, it is end-to-end and reproducible: data cleaning, evaluation, fairness analysis, and report assets are all scripted. Second, it compares methods of very different complexity under a common ranking protocol. Third, it includes interpretability and robustness analyses rather than only point metrics. Fourth, it uses a local LLM deployment path, which is practically relevant for privacy-sensitive HR settings.

9.2 Limitations

Four limitations are especially important and should shape interpretation of all results.

- **Proposal targets were not achieved.** The original proposal aimed for substantially higher Precision@10 and MRR than the final benchmark delivered. The value of the project therefore lies more in the comparative

methodology and the deployment-oriented interpretation than in hitting the initial numeric targets.

- **Weakly supervised label construction.** The benchmark labels are algorithmically constructed from category rules and overlap scores, not from recruiter decisions. This likely advantages overlap-sensitive models such as TF-IDF and may understate the value of richer semantic models on real human-labeled data.
- **LLM full-test run uses a compact prompt configuration.** To make the 823-job evaluation tractable on a local model, the final full-test LLM pass uses shorter text snippets and concise JSON outputs without rich explanation text. The ranking comparison is therefore stronger than before, but the explanation layer is thinner.
- **LLM fairness only smoke-tested.** Fairness for the LLM was audited on one job only. Any bias conclusion for the LLM would be premature.

Additional caveats also matter. The candidate pools are limited by split-wise resume partitioning, category mapping is heuristic, skill extraction is regex-based, and runtime results are partly environment-specific.

9.3 Business implications

From a business analytics perspective, the strongest deployment takeaway is that **a transparent lexical baseline remains highly competitive**. Under this benchmark, TF-IDF is not only the best full-test model but also the most stable fairness-wise and one of the cheapest to run. For organizations needing a first production baseline, TF-IDF offers a favorable quality–cost–auditability trade-off.

The **LLM reranker** is attractive for downstream refinement of a small shortlist, especially when recruiter-facing rationale text is valuable. However, its current cost profile and limited validation make it unsuitable as a full replacement for baseline screening. A more realistic production design would be a staged system: lexical retrieval for high recall and low cost, followed by selective reranking only for borderline or high-value cases.

The **SBERT result** is a useful negative finding. Dense semantic retrieval did not outperform the lexical baseline here, and it was also less robust to demographic perturbations. This does not mean dense models are always inferior; rather, it suggests that under weak supervision and category-heavy data, semantic generalization alone is not enough.

9.4 Recommended production posture

Based on the current evidence, the safest production posture is conservative. A practical system could use TF-IDF

as the default first-stage retriever, expose lexical overlap evidence for recruiter audit, and reserve LLM reranking for selected borderline or high-value cases. Before any broader rollout, the organization should collect human relevance judgments, expand the LLM run to the full test split, and perform a materially larger fairness audit.

An equally important process implication is governance. Because screening recommendations can affect livelihoods, every deployment should log model version, candidate lists, explanation traces, and recruiter overrides. The current codebase already moves toward that standard by saving ranking files, LLM traces, and fairness artifacts.

10 Conclusion

This project benchmarked TF-IDF, SBERT, and LLM-assisted reranking for resume–job matching under a reproducible pipeline with interpretability and fairness checks. The strongest empirically supported conclusion is that TF-IDF is the most reliable overall method on the current benchmark: it wins on the full held-out test split, is statistically better than SBERT, remains slightly ahead of the LLM on key metrics, runs cheaply, and is most robust under perturbations. The full-test LLM run strengthens the case for selective reranking because it now clearly beats SBERT, but it still does not justify replacing TF-IDF as the default baseline. Future work should replace weak labels with recruiter-validated judgments and perform a materially larger fairness audit before any deployment claim is made.

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